Thermodynamic description of the Bi–Sn system: a synthetic benchmark document

ORCHID synthetic benchmark (generated; no human authors)

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Abstract

This synthetic document describes the Bi–Sn binary system as calculated with pycalphad from the SGTE dataset BiSn-08Li.tdb; every figure, table and stated value is generated, and the exact values are kept in a separate truth key. The highest invariant reaction, L → (Bi) + (Sn), lies at 415.8 K.

1. Introduction

The Bi–Sn system belongs to the 20 binaries of the ORCHID benchmark, chosen to cover diagram types such as simple eutectic. The thermodynamic description is the SGTE collection entry [1], which reproduces the assessment of Z. Li, Z. Cao, S. Knott, A. Mikula, Y. Du, Z. Qiao, Calphad, 32, 152-63(2008). [2]. The calculations use pycalphad 0.11.2 at 1 atm.

Pure Bi melts at 544.5 K. For Sn, the liquidus meets the axis at 505.1 K.

2. Phase diagram and invariant reactions

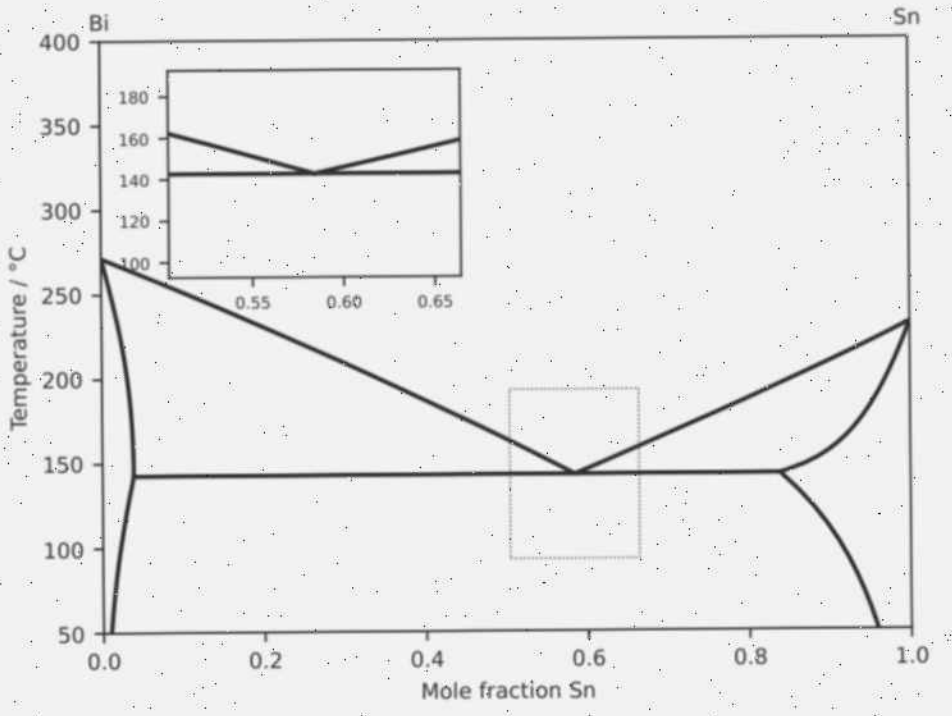


Fig. 1. Calculated Bi–Sn phase diagram (inset: enlarged view near the lowest invariant).

Eutectic equilibrium L → (Bi) + (Sn) is found at 415.8 K and 58.4 at.% (L).

Table 1 compares the calculated temperatures with experimental values from the

literature [3–5]; in this synthetic document those values and references are fictitious.

Table 1. Invariant reactions and congruent points, calculated and experimental (Bi–Sn).

Reaction	Type	T (°C) this work	T exp.[b]	Ref.	x(Sn), 1	x(Sn), 2	x(Sn), 3
L → (Bi) + (Sn)	eutectic	142.7	137.7	[5]	0.584	0.038	0.838

Phases 1–3 in the order written in the reaction. Calculated with pycalphad 0.11.2 from the SGTE dataset; synthetic document.

[b] Experimental temperatures are fictitious values attributed to invented references [3–5] for testing evidence-class extraction; they are not data.

3. Phase stability

Table 2. Solid phases: structure, stability range and homogeneity range (Bi–Sn).

Phase	Pearson	SB	Prototype	T min (°C)	T max (°C)	Range (at.% Sn)	T (°C)
(Bi)	hR2	A7	α-As	< 27	271.4	0.0–3.8	147
(Sn)	tI4	A5	β-Sn	< 27	231.9	85.1–100.0	137

The upper stability limit of (Bi) is 544.5 K. The solubility of Sn in (Bi) reaches 3.8 at.% at 420.0 K.

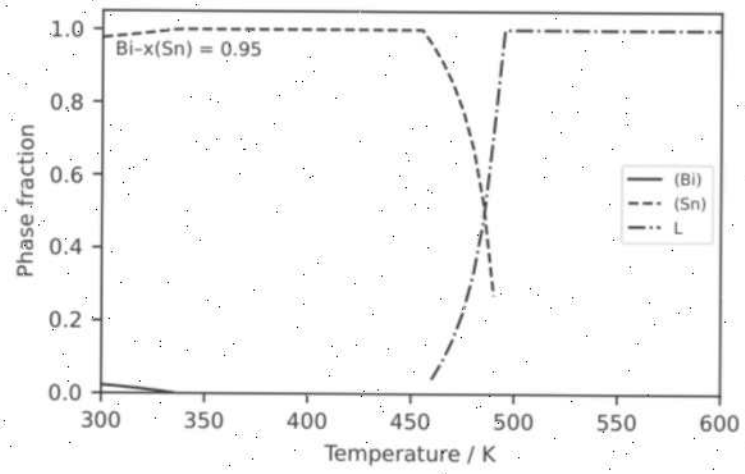


Fig. 2. Calculated equilibrium phase fractions versus temperature for Bi–x(Sn) = 0.95.

4. Thermodynamic properties

At 650 K the integral mixing enthalpy of the liquid has its maximum, 121 J/mol, at 64.0 at.% Sn. For the equiatomic liquid at 650 K, the Gibbs energy of mixing is −3.42 kJ/mol-atom.

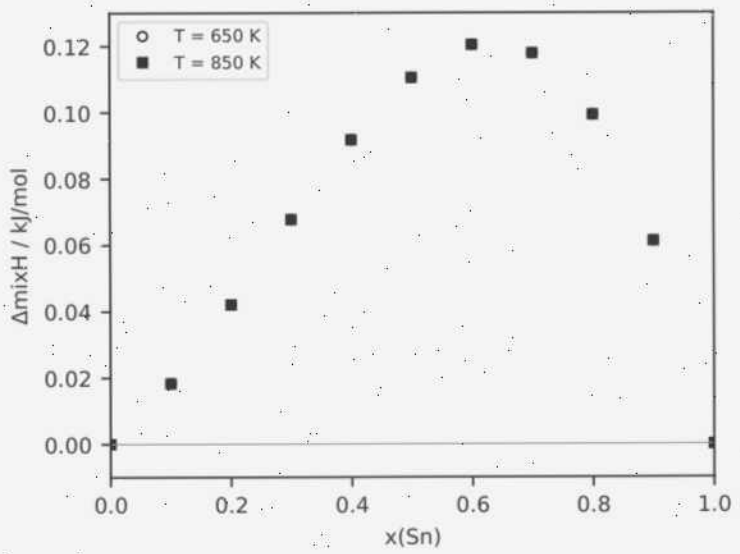


Fig. 3. Calculated mixing enthalpy of the liquid at 650 K and 850 K.

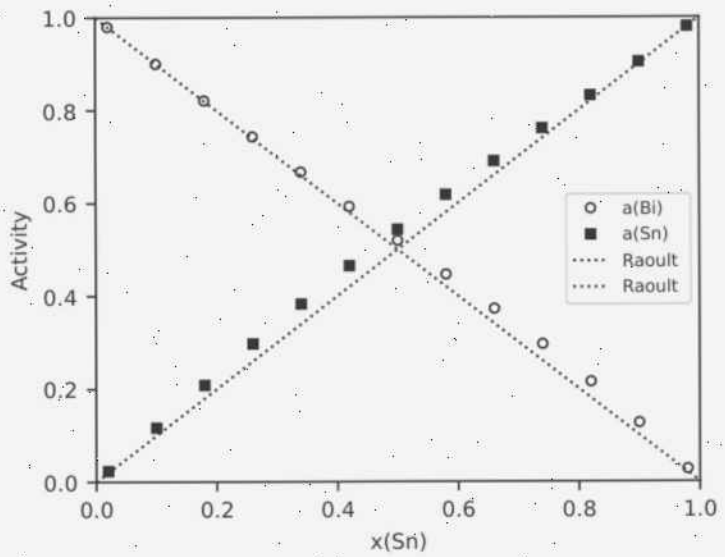


Fig. 4. Calculated activities in the liquid at 650 K (reference: pure liquids).

Table 3. Redlich–Kister parameters of the liquid (Bi–Sn).

ν	a / J mol⁻¹	b / J mol⁻¹ K⁻¹	c / J mol⁻¹ K⁻¹
L0	441.8	1.33493	0.0
L1	-298.1	-0.28229	0.0

Values as printed in the TDB file.

Table 4. Integral and partial mixing properties of the liquid at 650 K (Bi–Sn).

x(Sn)	ΔmixH / kJ mol⁻¹	ΔmixG / kJ mol⁻¹	a(Bi)	a(Sn)	ΔH(Bi) / kJ mol⁻¹	ΔH(Sn) / kJ mol⁻¹
0.100	0.02	-1.67	0.9001	0.1165	0.00	0.21
0.200	0.04	-2.54	0.8015	0.2309	-0.01	0.24
0.300	0.07	-3.07	0.7052	0.3408	-0.01	0.25
0.400	0.09	-3.35	0.6114	0.4449	0.00	0.22
0.500	0.11	-3.42	0.5195	0.5432	0.04	0.18
0.600	0.12	-3.30	0.4281	0.6363	0.09	0.14
0.700	0.12	-2.99	0.3349	0.7258	0.19	0.09
0.800	0.10	-2.45	0.2362	0.8142	0.32	0.04
0.900	0.06	-1.60	0.1271	0.9043	0.50	0.01

Reference states: pure liquid Bi and Sn at 650 K; per mole of atoms. ΔH: partial molar enthalpies of mixing.

5. Conclusions

This document is synthetic and exists only to test extraction tools; its truth key lists every boundary, field, invariant, plotted point, table cell and stated value.

References

[1] B. Hallstedt, SGTE binary dataset collection, Calphad 89 (2025) 102833 (CC BY 4.0).

[2] Z. Li, Z. Cao, S. Knott, A. Mikula, Y. Du, Z. Qiao, Calphad, 32, 152-63(2008).

[3] K. Synthetica, L. Exemplar, J. Fict. Phase Equilib. 12 (1987) 345–352 — fictitious reference, invented for this synthetic benchmark.

[4] M. Placeholder, Z. Notreal, Fict. Thermochim. Acta 41 (1994) 77–89 — fictitious reference, invented for this synthetic benchmark.

[5] R. Imaginary, Proc. Synth. Calorim. Soc. 8 (2003) 1102–1110 — fictitious reference, invented for this synthetic benchmark.